Wenjing Gong

Ph.D. student
Urban and Regional Science
College of Architecture
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(Updated on 11/2026)

RESEARCH & TEACHING INTERESTS

- Climate and Environmental Hazards, Extreme Heat, Air Pollution and Wildfire Smoke, Compound Hazards
- Human Mobility and Accessibility, Climate Adaptation and Resilience, Environmental Health and Justice
- Urban Analytics, Urban Planning, GIScience, GeoAI, Spatiotemporal Modeling, Urban Digital Twins

EDUCATION

2024 – 2028 **Ph.D. Student, Urban and Regional Science (Urban Analytics & GeoAI)**
Chair: Dr. Lingqian (Ivy) Hu, Co-Chair: Dr. Xinyue Ye
Graduate Certificate in Geographic Information Science (Department of Geography)
Texas A&M University, College Station, USA

2019 – 2022 **M.E., Architecture (Urban Study)**
Tongji University, Shanghai, China
GPA: 88.5/100 (Entrance through the exam-free recommendation program)

2019 **Visiting Student, Graduate School of Architecture, Planning and Preservation**
Columbia University, New York, USA

2014 – 2019 **B.E., Architecture**
Shandong University of Science & Technology, Qingdao, China
GPA: 90.5/100 (Rank: 1/60 each year)

ACADEMIC & PROFESSIONAL EXPERIENCE

2025 – pres. **Graduate Teaching Assistant**
Instructor: Dr. Douglas F. Wunneburger, Texas A&M University, College Station, USA

2025 – 2026 **Student Ambassador**
Texas A&M Institute of Data Science, College Station, USA

2025 – 2026 **Student Fellow**
Telehealth Institute, Texas A&M University, College Station, USA

2024 – 2025 **Graduate Research Assistant**
Advisor: Dr. Xinyue Ye, Center for Geospatial Sciences, Applications and Technology, Texas A&M University, College Station, USA

2022 – 2024 **Designer and Planner**

Architecture Design & Research Institute of Tongji University, Shanghai, China

AWARDS & HONORS

External

2026 Environmental Perception and Behavior Geography (EPBG) Travel Grant, 2026 Annual Meeting of American Association of Geographers, $150.

2026 International Geographic Information Fund (IGIF) Scholarship Award, 2026 Annual Meeting of American Association of Geographers, $1,200.

2026 Applied Geography Specialty Group Annual Meeting Award, 2026 Annual Meeting of American Association of Geographers, $202.

2026 Second Place, Geographic Information Science and Systems Specialty Group (GISSG) Student Honors Paper Competition (Third Author), 2026 Annual Meeting of American Association of Geographers.

2026 2025–2026 NSF I-GUIDE Spatial AI Challenge — GeoAgent4Disaster (Accepted Project).

2025 Travel Grant, Summer School on Cyberinfrastructure and Disaster Resilience, TAMU (funded by NSF), $200.

2025 Ashok K. Dutt Award for Best Graduate Student Paper (1st place), Regional Development and Planning Specialty Group, 2025 Annual Meeting of American Association of Geographers, $200.

2025 UCGIS I-GUIDE 2024–2025 Community Champions, $3,000 (Student Lead; PI: Dr. Xinyue Ye).

2025 NVIDIA Support for Short-Term Compute Needs on "using digital twins and transformer networks to model & mitigate human heat stress".

2019 "Outstanding Graduate" of Shandong Province, China (5%).

2018 National Best Leader Award, National Green Building Design Competition.

2017 Excellence Award, China Architectural Education/TSINGRUN Award Students' Paper Competition.

2016 – 2018 Multiple national architecture and design competition awards (**11 times**)

Internal

2026 Second Place (Jury Vote), Graduate Poster Session, COA AI Symposium, TAMU, up to $500.

2025 Texas A&M Institute of Data Science (TAMIDS) Student Ambassador Scholarship, Domain Data Science Track, TAMU, $2,000.

2024 – 2026 Travel Grant, LAUP Department, TAMU, $600 (**3 times**).

2024 – 2028 Graduate Assistantship covering stipend and tuition, Center for Geospatial Sciences, Applications and Technology, TAMU.

2019 – 2022 Academic Scholarship covering all tuition fees, TJU.

2014 – 2019 First-class Scholarships in academic performance (5%, **8 times**), SDUST.

2014 – 2019 First-class Corporate Scholarships (5%, **2 times**), SDUST.

2014 – 2019 "Outstanding Student Pioneer" (5%, **3 times**), SDUST.

2019 "Outstanding Student" in Science, Technology, and Innovation (5%), SDUST.

PUBLICATIONS

* Denotes my corresponding authorship, indicating a leading or supervisory role.

Selected Publications

2026 Huang, T., Ye, X., Zhai, W., Dong, X., Liu, C., Li, S., & **Gong, W.** (2026). AI-Driven Tree Placement Options for Heat Mitigation in Residential Yards. *Building and Environment*, 115301. https://doi.org/10.1016/j.buildenv.2026.115301

2026 Yang, Y., **Gong, W.**, Zhang, K., Zou, L., Tu, Z., Li, H., ... & Ye, X. (2026). RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. arXiv preprint arXiv:2606.21819. (Accepted in SIGSPATIAL 2026, full Applications Track paper)

2026 Rui, J., Shabrina, Z., & **Gong, W.*** (2026). Artificial intelligence applications in urban extreme heat management: A systematic review of forecasting, monitoring, mitigation and decision support. *Environmental Impact Assessment Review*, 119, 108363. https://doi.org/10.1016/j.eiar.2026.108363

2026 Ye, X., **Gong, W.**, Yang, Y., Zou, L., Tu, Z., Huang, X., ... & Wu, L. (2026). Towards Agentic Urban Digital Twins (AUDiTs): advancing new urban science through Human-AI co-learning agents. *Urban Informatics*, 5(1), 9. https://doi.org/10.1007/s44212-025-00099-3

2026 Rui, J., & **Gong, W.*** (2026). Social media-driven multi-scale flood sensing: A systematic review of applications from global monitoring to community resilience. *Journal of Environmental Management*, 407, 129895. https://doi.org/10.1016/j.jenvman.2026.129895

2026 **Gong, W.**, Wu, L., Zhu, C., Song, Y., & Ye, X. (2026). Revealing park visitation under dual environmental threats in a socially stratified city: Evidence from smartphone mobility data in Dallas. *Urban Forestry & Urban Greening*, 117, 129264. https://doi.org/10.1016/j.ufug.2026.129264

2025 **Gong, W.**, Ye, X., Wu, K., Jamonnak, S., Zhang, W., Yang, Y., & Huang, X. (2025). Integrating Spatiotemporal Vision Transformer into Digital Twins for High-Resolution Heat Stress Forecasting in Campus Environments. *Journal of Planning Education and Research*, 0739456X251391121. https://doi.org/10.1177/0739456X251391121

2025 Ye, X., Yigitcanlar, T., Xu, Y., Head, M., Hang, Y., Sanchez, T., **Gong, W.**, & Niyogi, D. (2025). Urban planning and IPCC-like city assessments integration for climate-resilient cities. *Environment and Planning B: Urban Analytics and City Science*, 23998083251330940. https://doi.org/10.1177/23998083251330940

2025 Ye, X., Yigitcanlar, T., Goodchild, M., Huang, X., Li, W., Shaw, S.-L., Fu, Y., **Gong, W.**, & Newman, G. (2025). Artificial intelligence in urban science: Why does it matter? *Annals of GIS*, 0(0), 1–9. https://doi.org/10.1080/19475683.2025.2469110

2024 **Gong, W.**, Rui, J., & Li, T. (2024). Deciphering urban bike-sharing patterns: An in-depth analysis of natural environment and visual quality in New York's Citi bike system. *Journal of Transport Geography*, 115, 103799. https://doi.org/10.1016/j.jtrangeo.2024.103799

2023 **Gong, W.**, Huang, X., White, M., & Langenheim, N. (2023). Walkability Perceptions and Gender Differences in Urban Fringe New Towns: A Case Study of Shanghai. *Land*, 12(7), 1339. https://doi.org/10.3390/land12071339

Additional Publications

2026 Zhou, S., Kodial, M., **Gong, W.**, Zhou, S., Newman, G., Li, X., ... & Ye, X. (2026). A pilot mobile application enhanced by augmented reality to support participatory CoDesign in shared public spaces. *Environment and Planning B: Urban Analytics and City Science*, 23998083261467083. https://doi.org/10.1177/23998083261467083

2026 Zhang, W., Zhai, W., Ye, X., **Gong, W.**, Yang, E., Bai, W., & Zhu, C. (2026). Planning and Computational Urban Research: A Comparative Analysis of Disciplinary and Interdisciplinary Agendas. *Applied Spatial Analysis and Policy*, 19(3), 145. https://doi.org/10.1007/s12061-026-09913-0

2026 Reed, K., Rybkowski, Z., Bardenhagen, E., **Gong, W.**, Starrett, C., Ennemoser, B., & Ye, X. (2026). Enhancing interdisciplinary competence in built environment research: Evaluating the impact of a two-day design charrette on undergraduates. *Frontiers of Urban and Rural Planning*, *4*(1), 2. https://doi.org/10.1007/s44243-025-00073-2

2026 Rui, J., & **Gong, W.** (2026). Paying lip service? An investigation into the spatial mismatch between younger and older adults' streetscape perceptual preference and visitation behavior. *Cities*, 171, 106750. https://doi.org/10.1016/j.cities.2025.106750

2025 Ye, X., Li, S., **Gong, W.***, Li, X., Li, X., Dadashova, B., Li, W., Du, J., & Wu, D. (2025). Street View Imagery in Traffic Crash and Road Safety Analysis: A Review. *Applied Spatial Analysis and Policy*, 18(2), 50. https://doi.org/10.1007/s12061-025-09653-7

2025 **Gong, W.**, Lee, C. S., Li, S., Adkison, D., Li, N., Wu, L., & Ye, X. (2025). Cyber victimization in hybrid space: An analysis of employment scams using natural language processing and machine learning models. *Journal of Crime and Justice*, 1–22. https://doi.org/10.1080/0735648X.2024.2448804

2021 Zhang, X., **Gong, W.** Evolution of Fire Administration Space: Analysis of Location and Architectural Form of the Fire Stations in the Concession Areas of Modern Shanghai. *Time+Architecture* (in Chinese), 182 (6), 156-165.

TEACHING EXPERIENCE

Guest Lecturer

Year	Term	Lecture Title	Course	Institution
2026	Fall	3D Tools in ArcGIS Pro: Mapping, Measuring, and Modeling the Vertical City	PLAN 625/URPN 325: GIS in Landscape and Urban Planning	TAMU
2026	Fall	Demographic Analysis: Acquisition, Modeling, and Analysis of Census Data With R	PLAN 626/URPN 326: Advanced GIS	TAMU

Graduate Teaching Assistant

Year	Term	Course	Institution
2026	Fall	PLAN 626/URPN 326: Advanced GIS	TAMU
2026	Fall	PLAN 625/URPN 325: GIS in Landscape and Urban Planning (Online)	TAMU
2026	Spring	PLAN 626/URPN 326: Advanced GIS	TAMU
2026	Spring	PLAN 625/URPN 325: Introduction to GIS	TAMU
2026	Spring	PLAN 625/URPN 325: Introduction to GIS (Online)	TAMU
2025	Fall	PLAN 626/URPN 326: Advanced GIS	TAMU
2025	Fall	PLAN 625/URPN 325: Introduction to GIS (Online)	TAMU
2020	Fall	Design Studio III (Urban Planning)	TJU

PRESENTATIONS

Conferences

2026 **Gong, W.** Hyperlocal $PM_{2.5}$ Inference with Low-Cost Sensors and Geospatial Foundation Model Embeddings for Environmental Hazard Assessment. *2026 American Geophysical Union*, December 7-11, 2026, San Francisco, California, USA (Poster).

2026 **Gong, W.** Mapping Extreme Heat Exposure and Environmental Deprivation in Sub-Saharan African Slums. *2026 American Geophysical Union*, December 7-11, 2026, San Francisco, California, USA (Poster).

2026 **Gong, W.** Toward Climate-Responsive Campuses: A Digital Twin Framework for High-Resolution Heat Stress Forecasting and Planning. *2026 Association of Collegiate Schools of Planning*, October 8-10, 2026, Pittsburgh, Pennsylvania, USA (Oral).

2026 Zhang, K., **Gong, W.** Built Environment Disparities in Post-Wildfire Recovery: Evidence from California Using Cross-View Imagery and Large Language Models. *2026 Association of Collegiate Schools of Planning*, October 8-10, 2026, Pittsburgh, Pennsylvania, USA (Accepted).

2026 **Gong, W.** Forecasting Heat Stress with AI in Digital Twins. *2026 Texas A&M College of Architecture AI Symposium*, October 5–6, 2026, College Station, Texas, USA (Poster).

2026 **Gong, W.** Hyperlocal $PM_{2.5}$ Imputation from Sparse Sensors with Earth Embeddings and State Space-Enhanced Spatial Learning. *2026 International Symposium on Spatiotemporal Data Science*, August 7-8, 2026, Virginia Tech, Virginia, USA (Oral).

2026 **Gong, W.**, Ye, X., Song, Y. Essential Infrastructure or Unavoidable Risk? Modeling Urban Park Visitation under Dual Climate and Crime Stressors with Large-Scale Mobility Data. *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA (Oral).

2026 Yang, Y., Zou, L., **Gong, W.**, Fu, K., Li, Z., Wang, S., Zhou, B., Cai, H., Tian, H. DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA (Oral).

2025 **Gong, W.**, Ye, X., Wu, K., Jamonnak, S., Zhang, W., Yang, Y., Huang, X. Integrating Spatiotemporal Vision Transformer into Digital Twins for High-Resolution Heat Stress Forecasting in Campus Environments. *2025 Annual Meeting of American Association of Geographers*, March 24-28, 2025, Detroit, Michigan, USA (Oral).

2024 **Gong, W**., Yang, Y., Zhang, W., Wu, K., Ye, X., Huang, X. Integrating high-resolution simulations and transformer network predictions for mitigating human heat stress: A digital twin of a Texas campus. *2024 Texas Weather Conference*, November 9, 2024, College Station, Texas, USA (Oral).

2024 **Gong, W**., Yang, Y., Zhang, W., Wu, K., Ye, X., Huang, X. Integrating high-resolution simulations and transformer network predictions for mitigating human heat stress: A digital twin of a Texas campus. *College of Architecture's 26th Annual Showcase*, October 3, 2024, Texas A&M University, College Station, Texas, USA (Poster).

2024 **Gong, W**. Simulating and Predicting Spatial-Temporal Human Outdoor Heat Exposure: A Case Study on a Texas Campus. *The 2024 Symposium on Spatiotemporal Data Science: GeoAI for Social Sciences*, July 24, 2024, Virtual (Accepted).

Workshops

2025 Yang, Y., **Gong, W.**, Zhang, K. DisasterVLP: Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models. *Summer School on Cyberinfrastructure and Disaster Resilience*, August 15, 2025, Texas A&M University, College Station, USA (Oral).

2022 Gu, J., **Gong, W**., Zhang, X. Analysis of the Spatial Distribution of Express Courier Stations and Assessment of Site Selection Effectiveness. *DigitalFUTURES 12th Summer Workshops*, July 2, 2022, Virtual (Oral).

2019 **Gong, W**., Huang, H., Zhang, X. Undoing Sprawl: Urban Agriculture as a Social Innovation. *Resizing the Urban Form in the Era of "Negative Population Growth" Workshop*, December 9, 2019, Columbia University, New York City, USA (Oral).

SERVICES

Editorial Boards

- Managing Editor: Computational Urban Science

Invited Reviewer

- **GIScience and Geography:** International Journal of Geographical Information Science, Transactions in GIS, Journal of Transport Geography, Applied Geography, Journal of Geovisualization and Spatial Analysis, Geocarto International, Applied Spatial Analysis and Policy
- **Urban Planning and Urban Studies:** Journal of Planning Education and Research, Landscape and Urban Planning, Cities, Sustainable Cities and Society, Computational Urban Science, City and Environment Interactions, Sustainable Futures
- **Environmental Science and Climate:** Environmental Impact Assessment Review, Urban Climate, Environmental Monitoring and Assessment, Physics and Chemistry of the Earth, Frontiers in Environmental Science
- **Transportation:** Transportation Research Part C, Journal of Transport and Land Use, Transportation Planning and Technology
- **Multidisciplinary:** Technology in Society, Scientific Reports, Frontiers in Psychology

Conference Organizations

- Session Co-chair: Symposium on Human Dynamics Research: Extreme Weather, Human Responses, and Urban Resilience (I & II). *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA.
- Session Chair/Primary Organizer: Symposium on Human Dynamics Research: Urban Environmental Intelligence and Human-Climate Interactions (1 & 2). *2025 Annual Meeting of American Association of Geographers*, March 24-28, 2025, Detroit, Michigan, USA.

Professional Membership

- American Association of Geographers (AAG)
- Association of Collegiate Schools of Planning (ACSP)
- American Geophysical Union (AGU)
- American Planning Association (APA)

MEDIA COVERAGE

- TAMIDS Announces 2025-2026 Cohort of Student Ambassadors. https://tamids.tamu.edu/2025/07/18/tamids-announces-2025-2026-cohort-of-student-ambassadors/
- Throwing Shade at Heatwaves: Tracking campus "hot spots" could help students stay cool in the Texas heat. https://engineering.tamu.edu/news/2024/09/throwing-shade-at-heatwaves.html

SKILLS

- **Programming:** Python (Data Processing, Visualization, and Analysis), SQL, HTML
- **Machine Learning/Deep Learning:** Agentic AI, Computer Vision, Time series Forecasting, Tree models, Explainable AI

- **Spatial Data Analysis:** Google Earth Engine, ArcGIS Pro, QGIS, Geoda, MGWR
- **Urban Climate Simulation:** SOLWEIG, Envi-met
- **Design and Planning:** AutoCAD, SketchUp, Rhino, Photoshop, Illustrator, InDesign, Enscape, Lumion